

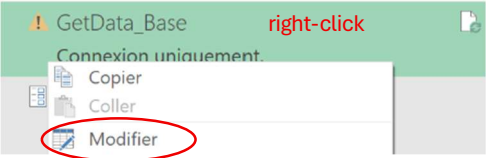
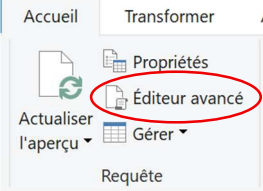
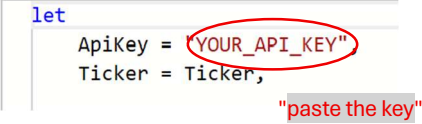
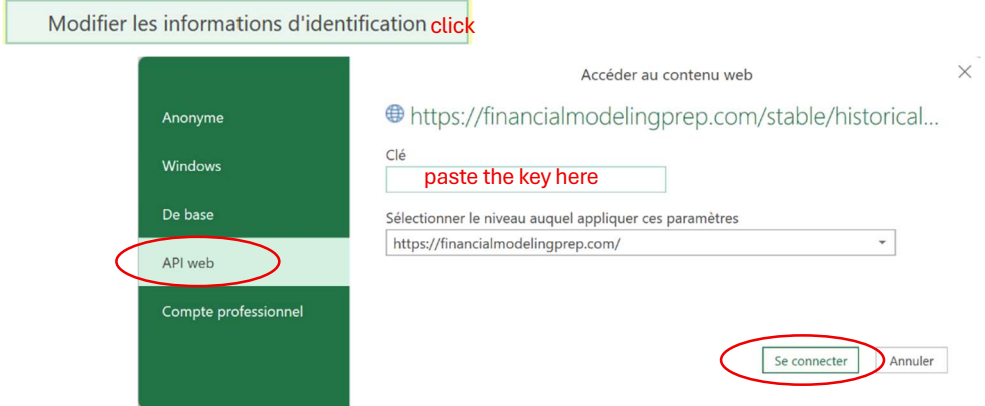
How to Use

Demo version | Last Updated: 30/09/2026 | by Ilona Gavoille

Requirements

To use this portfolio optimization model, you will need:

- **Microsoft Excel Desktop**
- **Solver Add-in** enabled in Excel
 - <https://support.microsoft.com/en-us/excel/load-the-solver-add-in-in-excel>
- **Internet connection** to retrieve financial market data
- **Personal API key** from the Financial Modeling Prep API:
 - Create an account to get an API key on <https://site.financialmodelingprep.com/>
 - Insert the key in Excel Query:

- 1. Excel ribbon with 'Données' tab selected.
- 2. Right-click context menu for 'GetData_Base' with 'Modifier' selected.
- 3. Excel ribbon with 'Éditeur avancé' selected.
- 4. Excel query editor showing 'let' statement with 'YOUR_API_KEY' placeholder. "paste the key"
- 5. Login dialog box for Financial Modeling Prep API. "Modifier les informations d'identification click" and "paste the key here".

1. Enable macros

When opening the file with format .xlsm file, **enable macros** if asked.

The workbook opens on the Dashboard, which is the main user interface.

2. Refresh the market data

Click **REFRESH**.

Note: The workbook initially contains data from the previous use. Always refresh the data before generating a new set of portfolios.

3. Set the optimization parameters

Once the data has been refreshed, click **OPTIMIZE PORTFOLIO** on the Dashboard.

Follow the guidelines and enter in the yellow cells:

- **Minimum weight** for each asset
- **Maximum weight** for each asset
- **Risk-free rate** : choose a rate appropriate for your analysis, such as the 10-Year U.S. Treasury yield, T-Bill rate, or another relevant risk-free benchmark.

These parameters define the constraints used by the optimization model.

4. Generate the portfolios

Once the parameters have been entered, click **GENERATE**.

The model automatically runs Excel Solver and generates:

- **12 portfolios** along the efficient frontier
- **1 Maximum Sharpe portfolio**
- **1 Minimum Variance portfolio**

The optimization may take a few moments depending on the number of constraints and the computer used.

The portfolio details, efficient frontier and Capital Allocation Line are displayed in the 03_Efficient Frontier sheet.

5. Explore the results

Return to the **Dashboard**. Use the **Selected Portfolio** dropdown to choose between the Portfolio 1 to Portfolio 12, Maximum Sharpe Portfolio or Minimum Variance Portfolio.

The Dashboard then updates dynamically according to the selected portfolio.